UVA 10128

Queue

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題意解釋

I. Every person has a different height
Can see P people, from the beginning
Can see R people, from the end

O. How many different permutations？

Sample Input

3 - Loop times

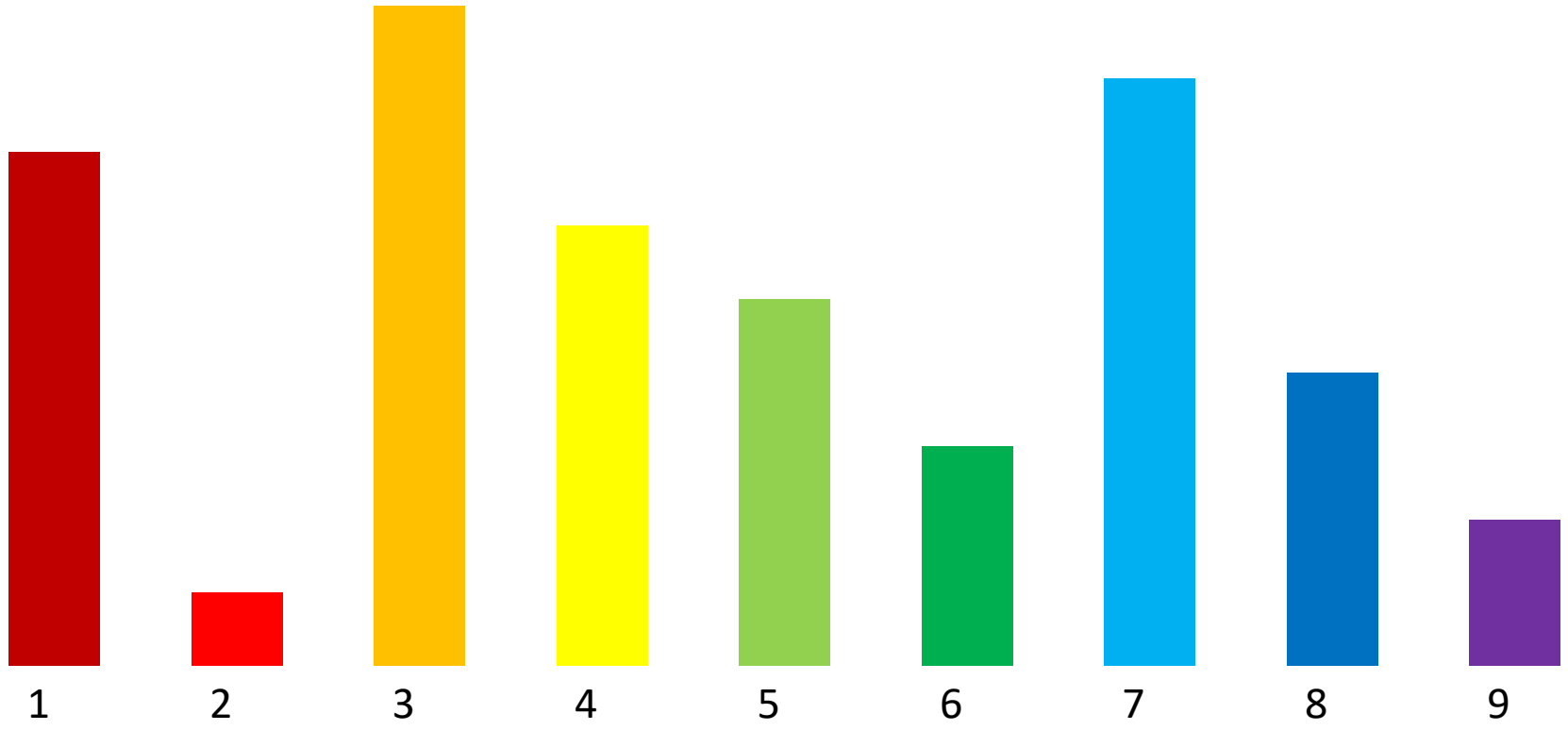
10 4 4 - N, P, R

11 3 1

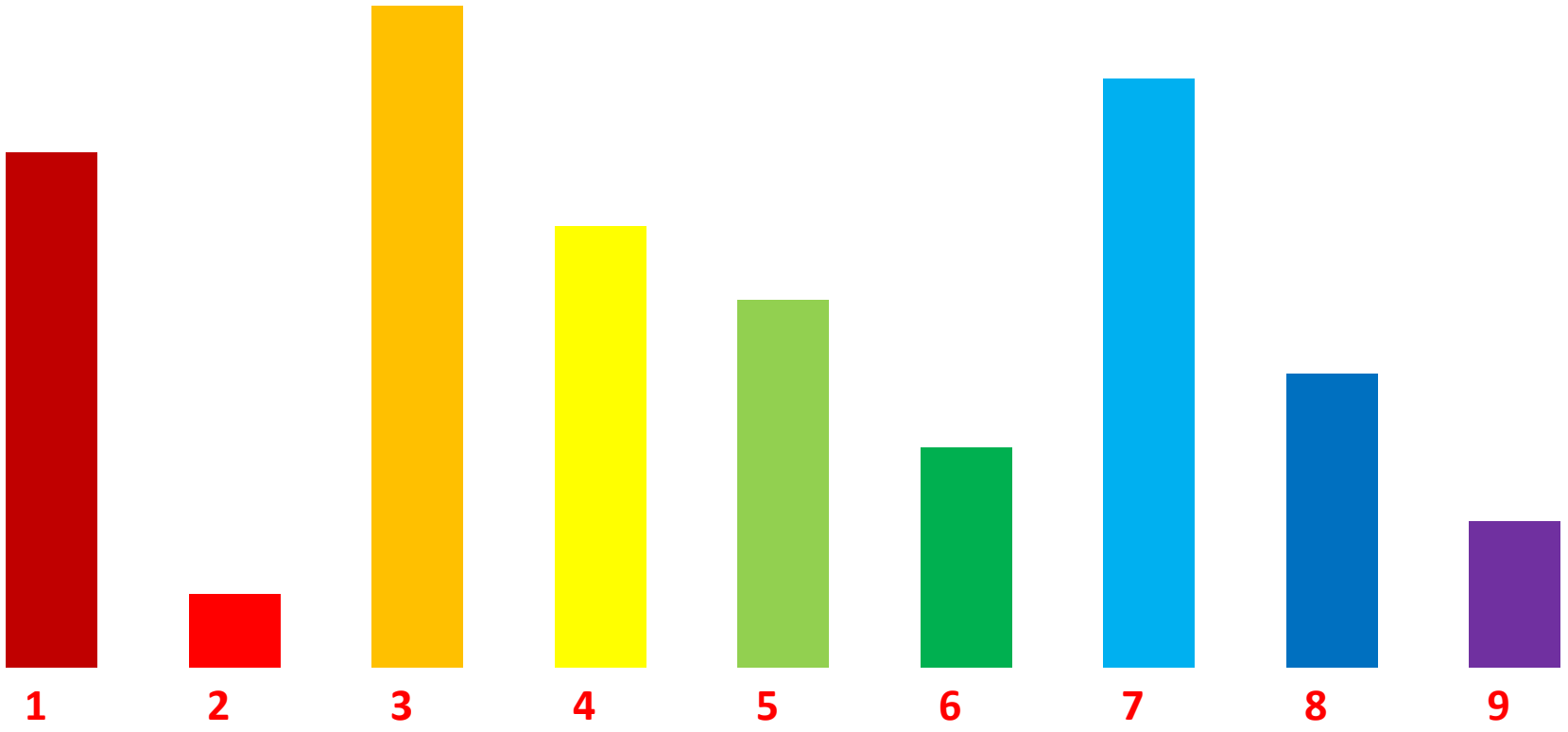
3 1 2

圖像化 ⟶

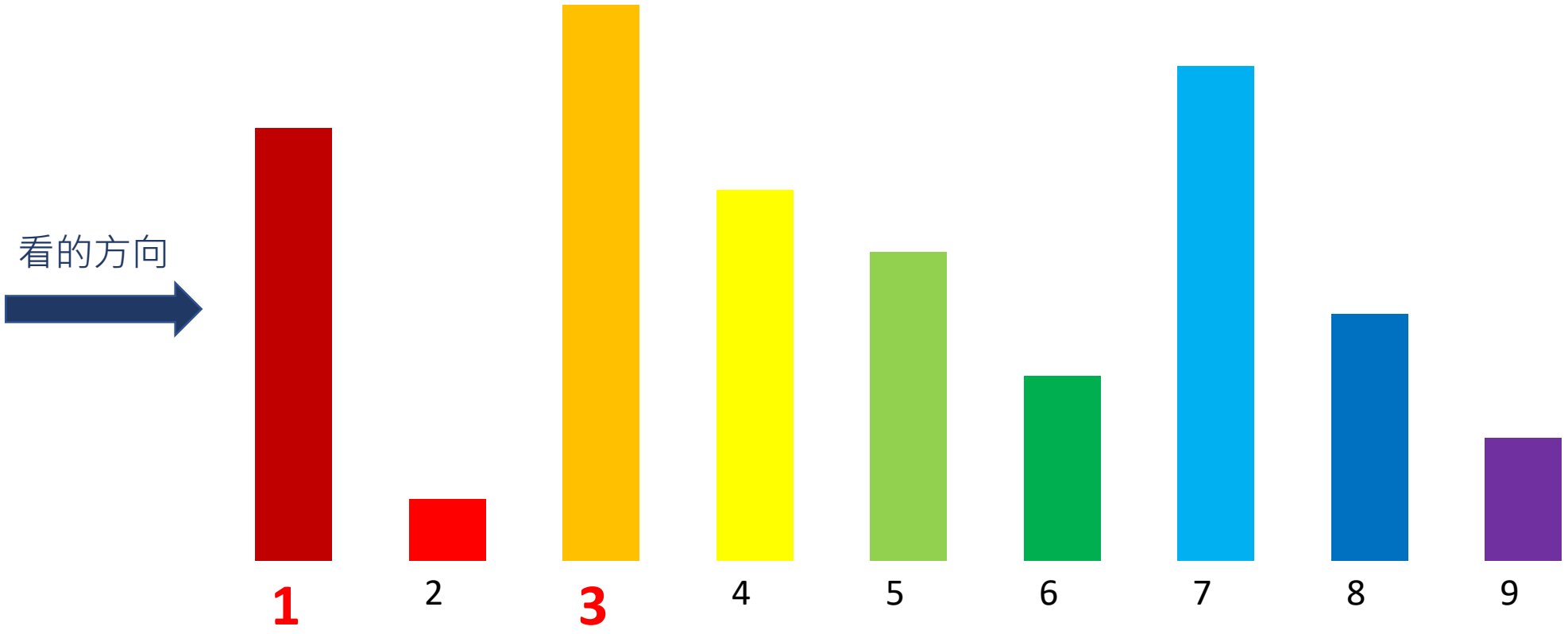
N = 9, P = 2, R = 4



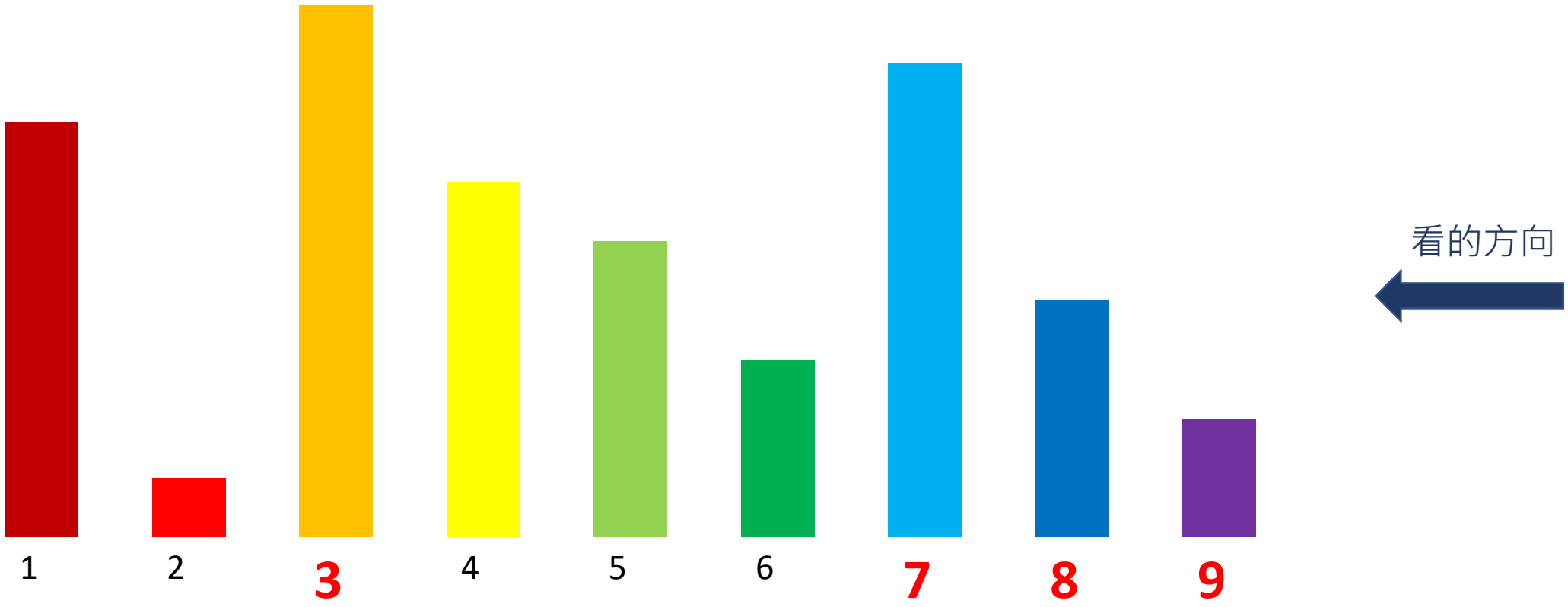
N = 9, P = 2, R = 4



N = 9, P = 2, R = 4



N = 9, P = 2, R = 4

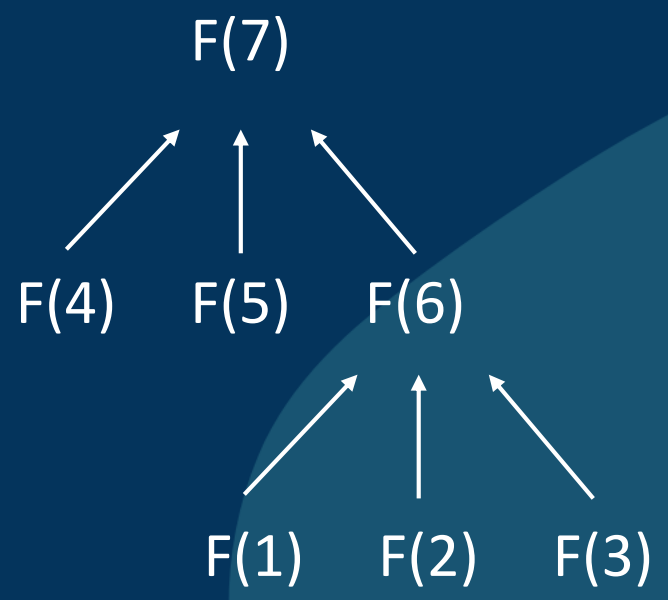


解題思路

Dynamic Programming

feature

- 分治法的延伸
- 將大問題切成小問題
- 用記憶法儲存這些問題的答案
- 避免重複求解
- 以空間換取時間



F(1)	F(2)	F(3)	F(4)	F(5)	F(6)	F(7)

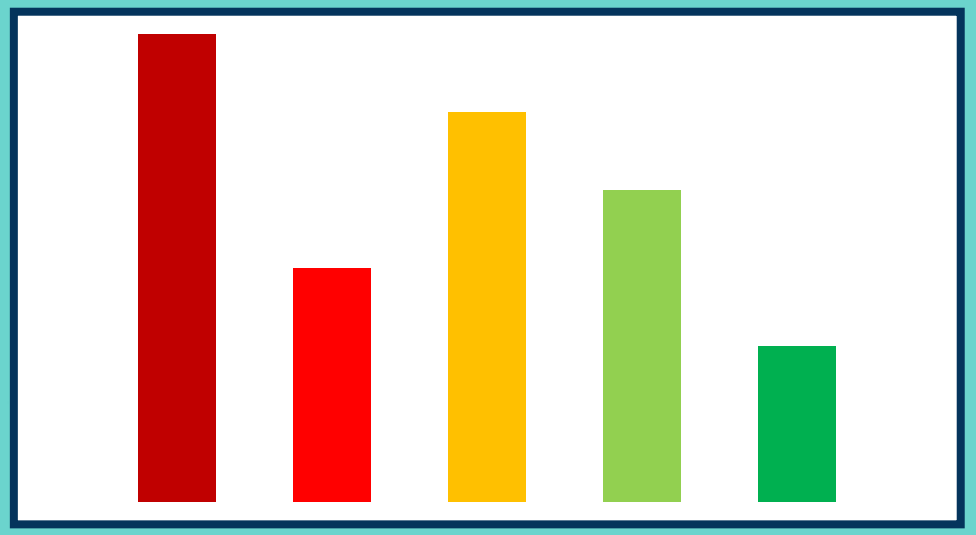
解題思路

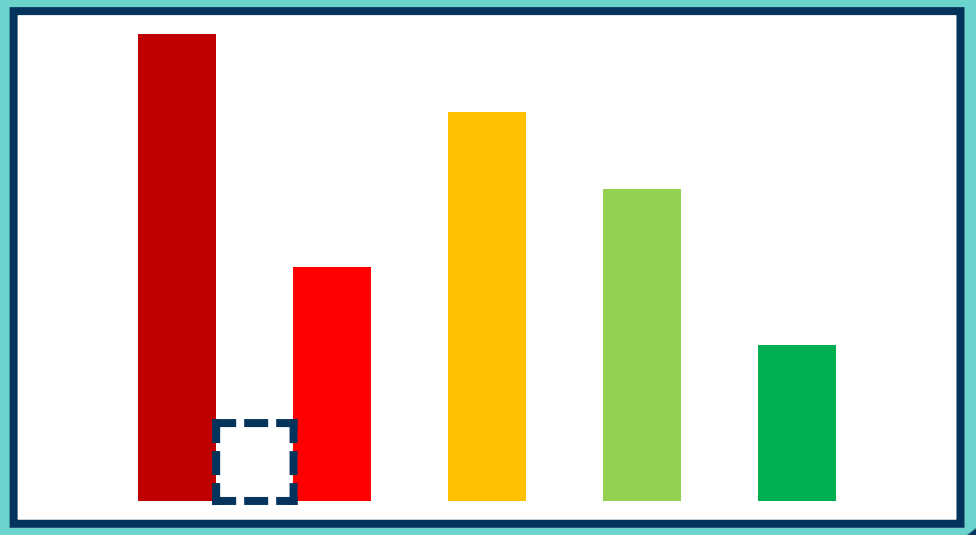
影響N-1
不影響P, R

影響N-1, P
不影響R

影響N-1, R
不影響P

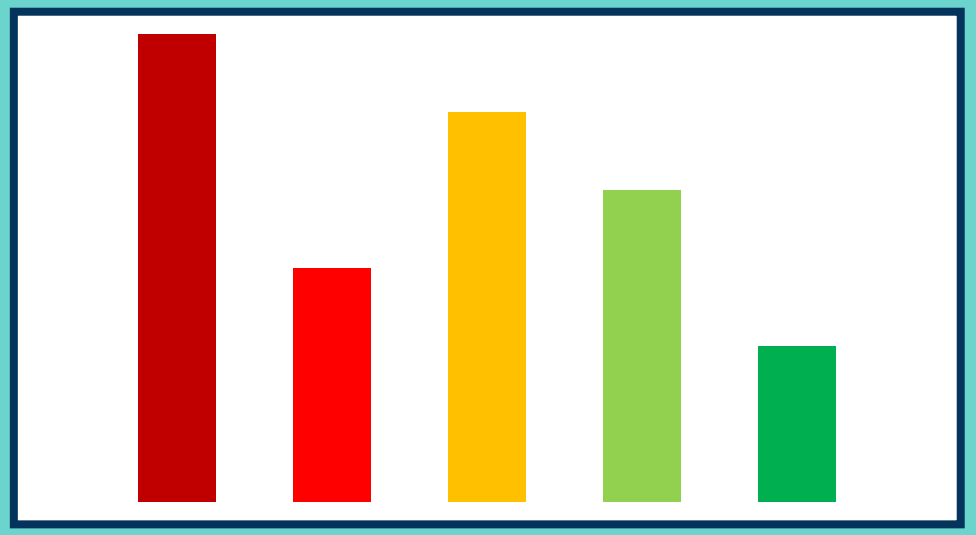
影響N-1 不影響P, R

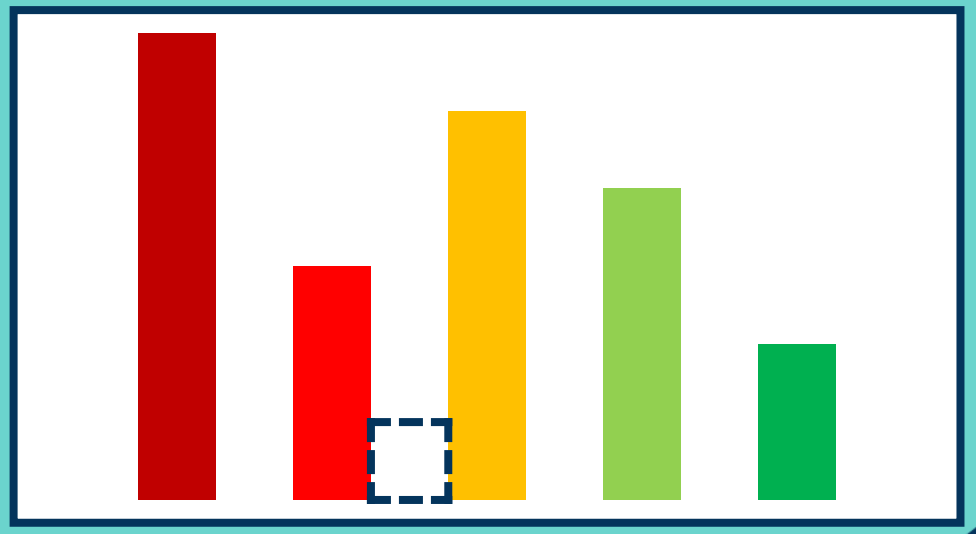




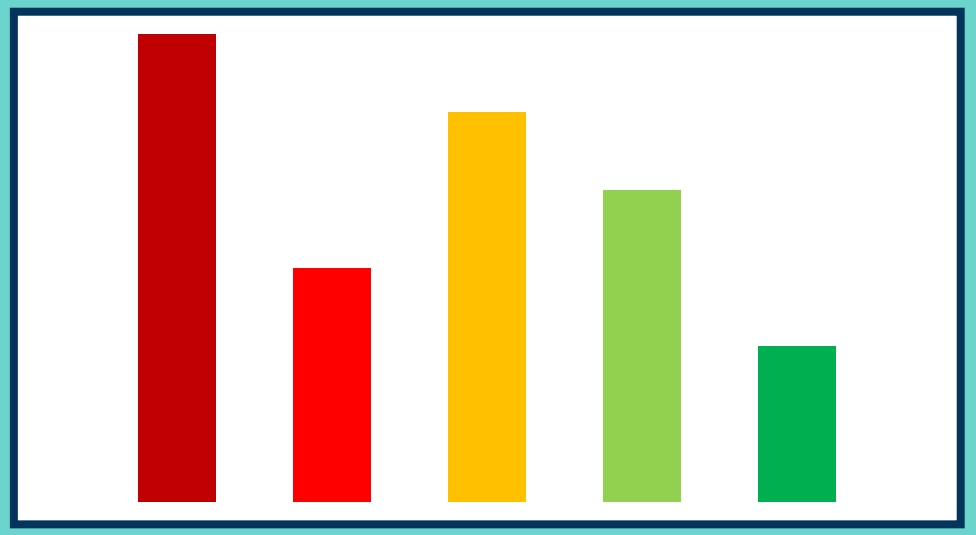
最矮的人原
本待的位置

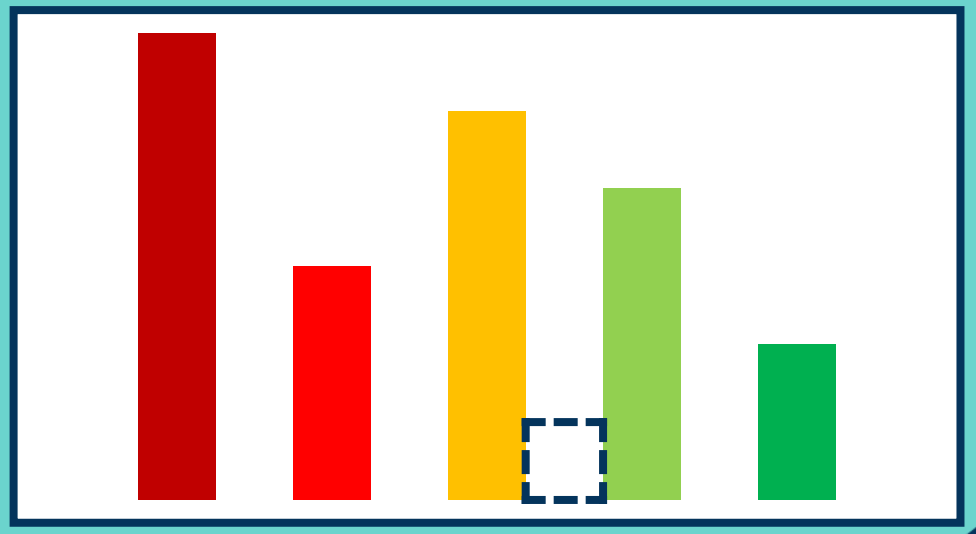
影響N-1 不影響P, R



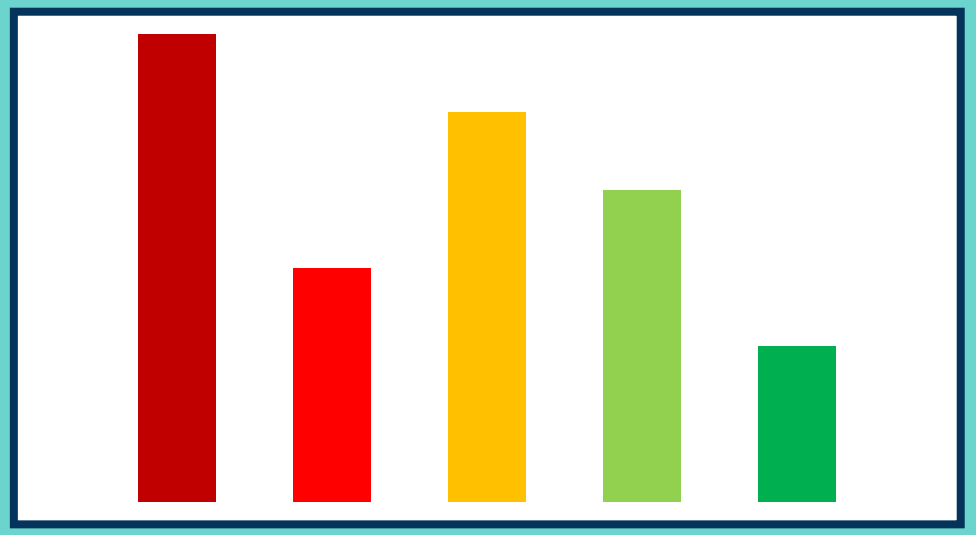


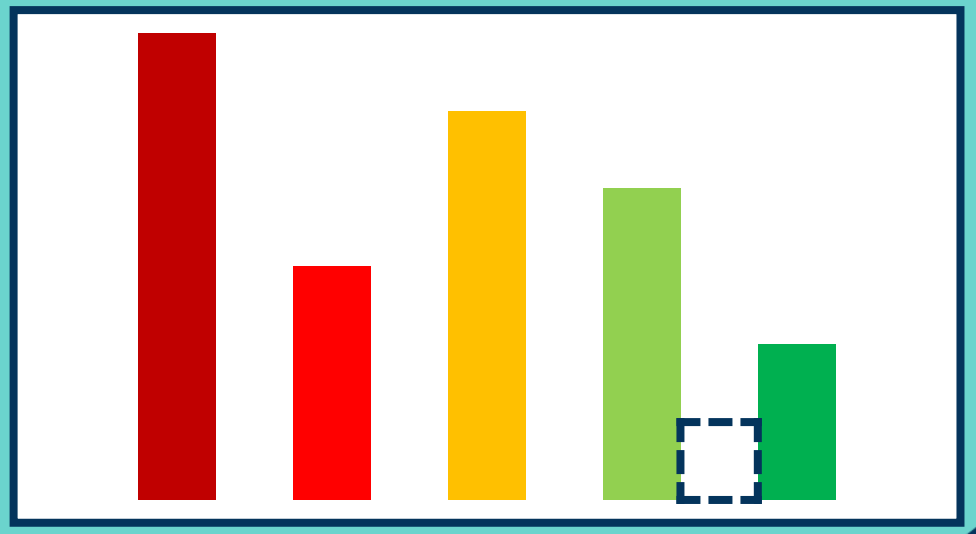
影響N-1 不影響P, R



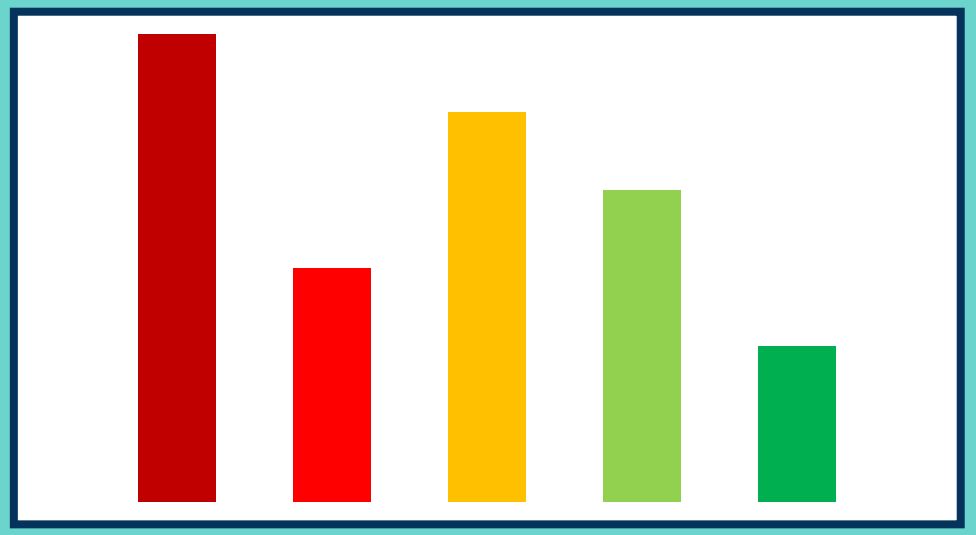


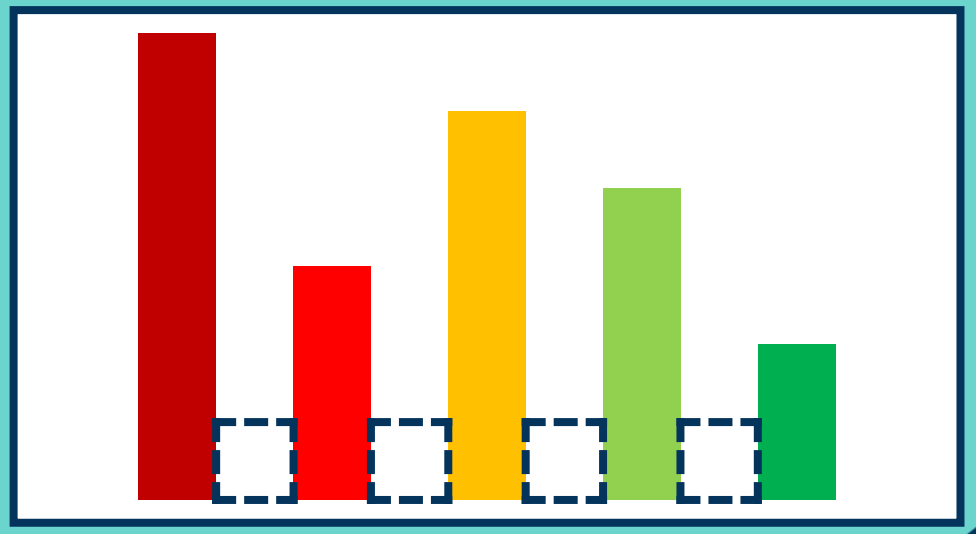
影響N-1 不影響P, R



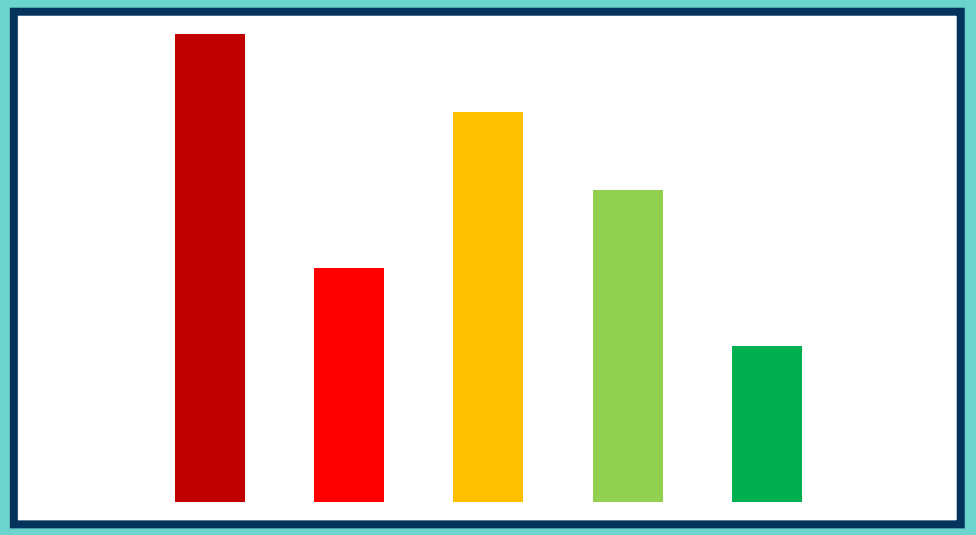


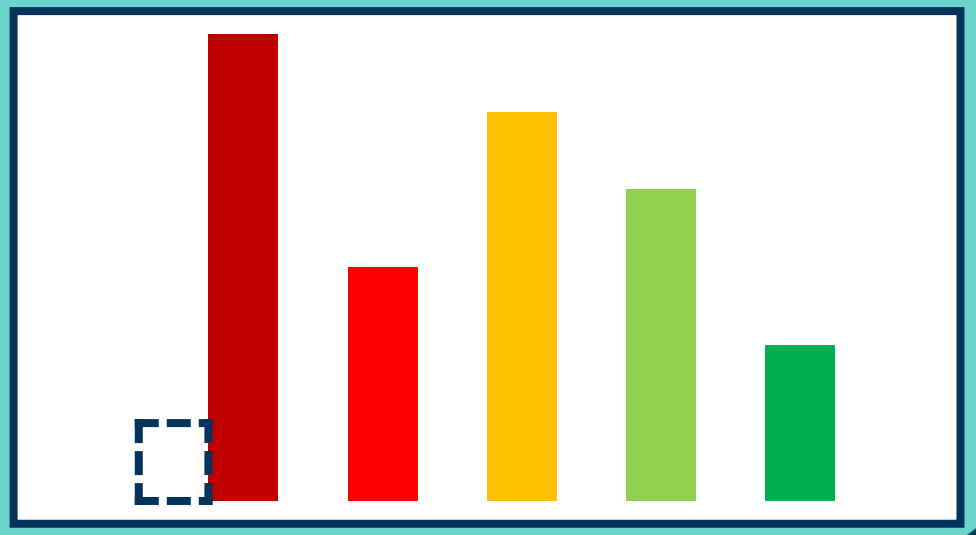
影響N-1 不影響P, R



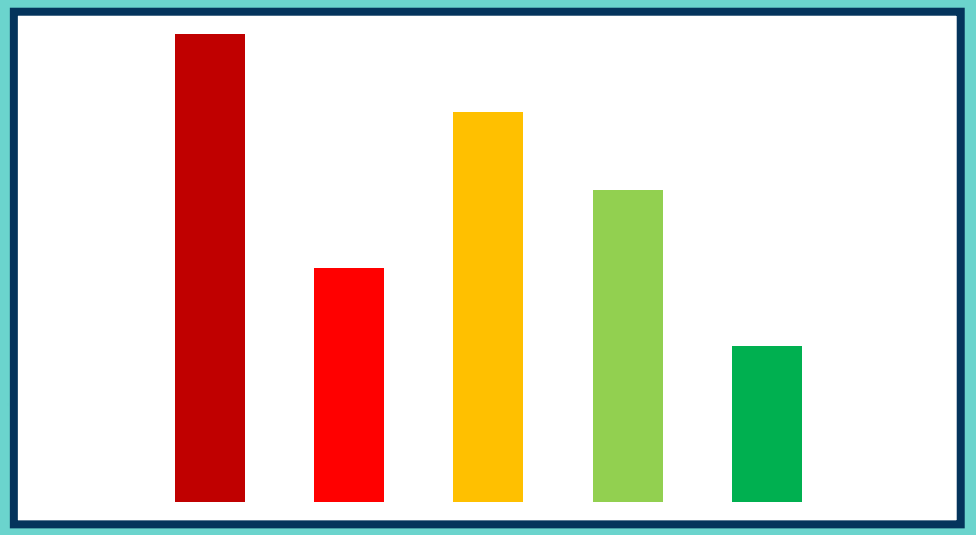


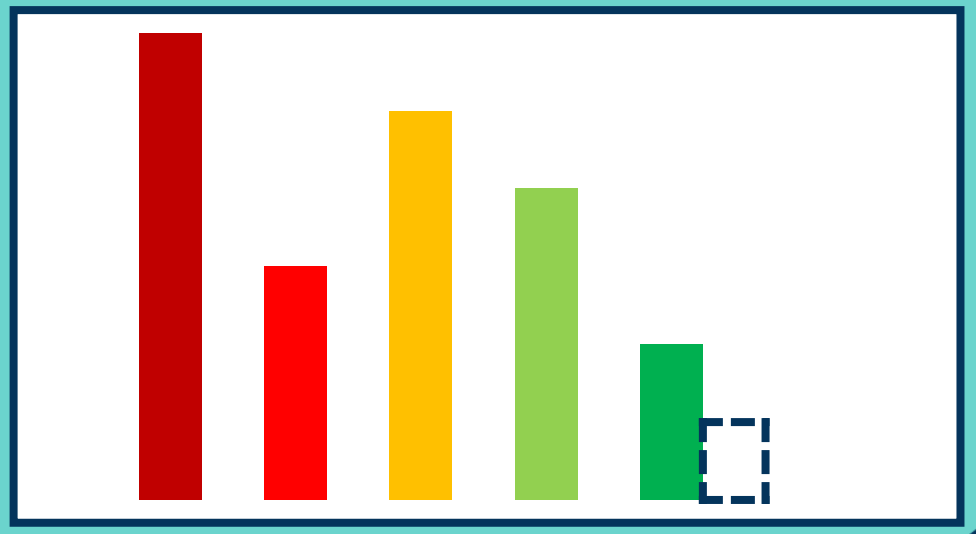
影響N-1, P 不影響P





影響N-1, R 不影響R





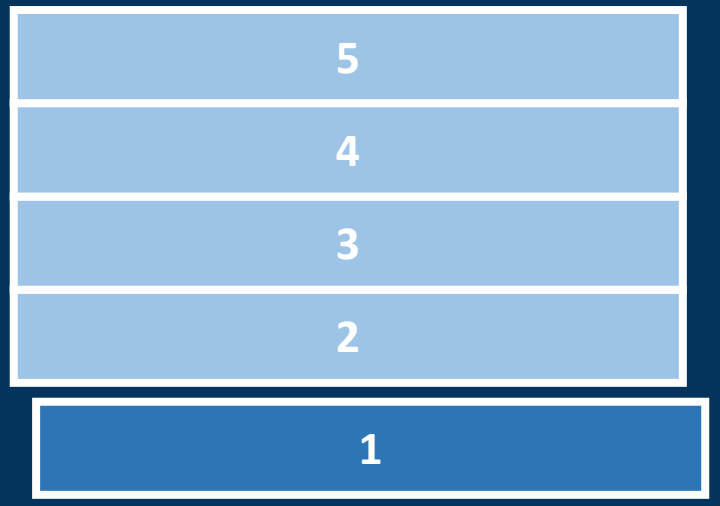
左 + 中 + 右

DP[n-1][p-1][r] + DP[n-1][p][r]*(n-2) + DP[n-1][p][r-1]

= DP[n][p][r]

DP[N][P][R] = DP[N-1][P][R]*(N-2) + DP[N-1][P-1][R] + DP[N-1][P][R-1]

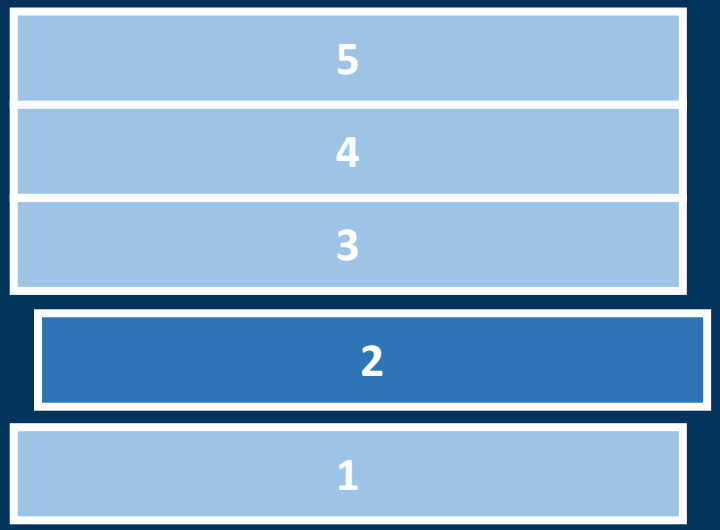
N = 1



	0	1	2	3	4	5	6	7	8	9	10	11	P
0	0	0	0	0	0	0	0	0	0	0	0	0	
1	0	1											
2	0												
3	0												
4	0												
5	0												
6	0												
7	0												
8	0												
9	0												
10	0												
11	0												
R													

DP[N][P][R] = DP[N-1][P][R]*(N-2) + DP[N-1][P-1][R] + DP[N-1][P][R-1]

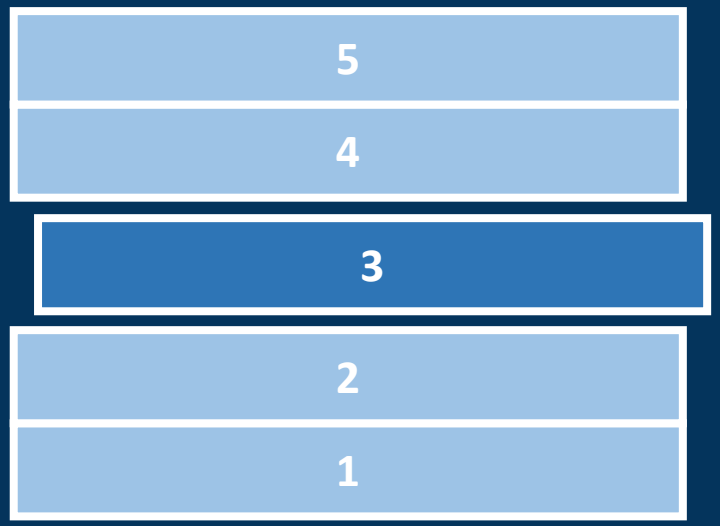
N = 2



	0	1	2	3	4	5	6	7	8	9	10	11	P
0	0	0	0	0	0	0	0	0	0	0	0	0	
1	0	0	1										
2	0	1											
3	0												
4	0												
5	0												
6	0												
7	0												
8	0												
9	0												
10	0												
11	0												
R													

$$DP[N][P][R] = DP[N-1][P][R]*(N-2) + DP[N-1][P-1][R] + DP[N-1][P][R-1]$$

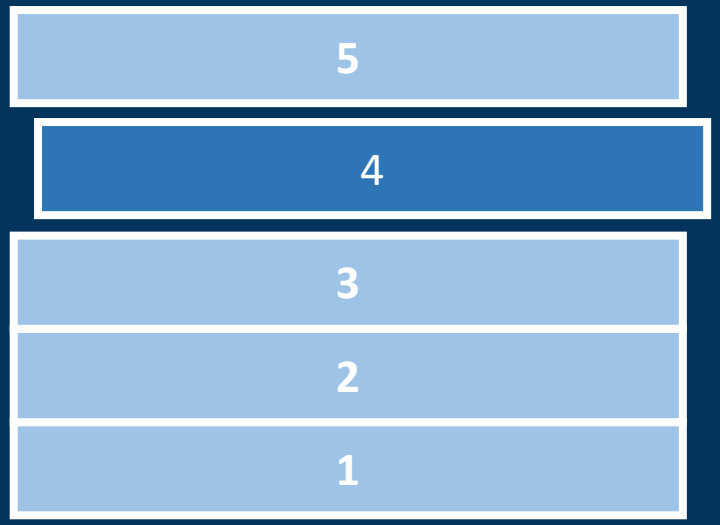
N = 3



	0	1	2	3	4	5	6	7	8	9	10	11	P
0	0	0	0	0	0	0	0	0	0	0	0	0	
1	0	0	1	1									
2	0	1	2										
3	0	1											
4	0												
5	0												
6	0												
7	0												
8	0												
9	0												
10	0												
11	0												
R													

DP[N][P][R] = DP[N-1][P][R]*(N-2) + DP[N-1][P-1][R] + DP[N-1][P][R-1]

N = 4



	0	1	2	3	4	5	6	7	8	9	10	11	P
0	0	0	0	0	0	0	0	0	0	0	0	0	
1	0	0	2	3	1								
2	0	2	6	3									
3	0	3	3										
4	0	1											
5	0												
6	0												
7	0												
8	0												
9	0												
10	0												
11	0												
R													

PSEUDO CODE

bigO(N^3)

```
input  N, P, R
DP[1][1][1] = 1
for I = 2 to 13
    for J = 1 to I
        for K = 1 to I
            DP[i][j][k] = DP[i-1][j][k]*(i-2)
            + DP[i-1][j-1][k] + DP[i-1][j][k-1];
Output  DP[N][P][R]
```

PYTHON CODE

```python
dp = [[[0 for _ in range(100)] for _ in range(100)] for _ in range(100)]
dp[1][1][1] = 1
for n in range(2,14):
    for p in range(1,n+1):
        for r in range(1,n+1):
            dp[n][p][r] = dp[n - 1][p - 1][r] + dp[n - 1][p][r - 1] + dp[n - 1][p][r]* (n - 2)
time = int(input())
while time>0 :
    N,P,R = map(int,input().split())
    print(dp[N][P][R])
    time=time-1
```

result

期末報告.py

C: > Users > j0918 > Desktop > 期末報告.py > ...

```python
dp = [[[0 for _ in range(100)] for _ in range(100)] for _ in range(100)]
dp[1][1][1] = 1
for n in range(2,14):
    for p in range(1,n+1):
        for r in range(1,n+1):
            dp[n][p][r] = dp[n - 1][p - 1][r] + dp[n - 1][p][r - 1] + dp[n - 1][p][r]* (n - 2)
time = int(input("輸入次數:"))
while time>0 :
    print("輸入測資:")
    N,P,R = map(int,input().split())
    print("輸出結果:")
    print(dp[N][P][R])
    print("-------------------------")
    time=time-1
```

問題 輸出 偵錯主控台 終端機 Python

```
輸入次數:3
輸入測資:
10 4 4
輸出結果:
90720
-------------------------
輸入測資:
11 3 1
輸出結果:
1026576
-------------------------
輸入測資:
3 1 2
輸出結果:
1
-------------------------
```

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9718992	f616773488 (isu11003022a)	d898. 10128 - Queue – UVa10128	AC (0.2s, 12.3MB)	PYTHON	2022-06-01 15:22
9714100	f616773488 (isu11003022a)	d898. 10128 - Queue – UVa10128	AC (3ms, 120KB)	CPP	2022-05-31 19:51

Thanks for listening